Deep Water AUV for Ocean Exploration Mission

1. Introduction and Purpose

The Office of Ocean Exploration and Research (OER) is seeking detailed information from potential providers regarding commercially available or near-production Deep Water Autonomous Underwater Vehicles (AUVs) that can support the OER ocean exploration mission. The information gathered will be used for internal planning and market analysis to determine the availability and suitability of commercial-off-the-shelf (COTS) or modified AUV systems that meet NOAA Ocean Exploration mission requirements. The target includes both Contractor-Owned/Contractor Operated (Data or Platform as a Service) responses and vendors intending to sell systems of platforms and sensors to the government as Government Owned-Government Operated or Government Owned - Contractor Operated business models.

2. Mission Requirement

OER is interested in AUV systems with the following preferred capabilities. Respondents are encouraged to describe systems that operate as close the capabilities as possible e.g. systems with operating depths of 4000m versus full ocean depth:

Feature	Preferred Capability
Maximum Operating Depth	6,000 meters or greater
Endurance	48 hours or greater
Navigation System	INS/DVL with Acoustic Positioning (USBL/LBL)
Maximum Speed	4 knots or greater
Payload Volume	1 cubic meter or greater
Communication	Real-time data telemetry (where possible)

Respondents are requested to describe the platform capabilities to sample a targeted subset of the 16 Priority Exploration variables described by NOAA *Ocean Exploration*, as categorized in the two groupings below (Table 1 and Table 2).

2.1 Core Variables and Capabilities (Table 1)

OER is interested in AUVs that can support the acquisition of, at a minimum, these core exploration variables of interest. OER is specifically interested in novel sensing approaches or systems combining sensors. Note that OER is interested primarily in the exploration variable, and that potential sensing approaches are given as indications or potential approaches. Not all sensing approaches are necessarily expected and descriptions of other sensing approaches to meet the exploration variables requirements are encouraged.

Exploration Variable of Interest	Potential Sensing Approaches for AUVs
Bathymetry and habitat types	Multibeam Echo Sounder (including backscatter), Side-scan or synthetic aperture sonar, Video/still imagery
Geology/mineral resource characterization	Sub bottom profiler, Video/still imagery / Very high-resolution bathymetry
Biomass, density, distribution, and abundance: *Microbes, plankton, meiofauna*	eDNA water samples, single beam echo sounder (e.g. EK80)
Biomass, density, distribution, and abundance: invertebrates, fish, megafauna, marine mammals	Video/still Imagery, eDNA water samples, single beam echo sounder (e.g. EK80)
Ocean physics and chemistry:	Conductivity Temperature Depth (CTD), Dissolved Oxygen (DO) sensor, pH sensor, Dissolved Inorganic Carbon (DIC) sensor, dissolved organic carbon, Oxygen Reduction Potential (ORP) sensor, Inorganic nutrient sensors (nitrate/nitrite, silicate, phosphate), Pressure sensor, Current measurement, Turbidity measurement, Particulate organic matter, Flow through imaging (not all sensing approaches are required or expected)

2.2. Stretch Goals for Data Acquisitions (Table 2)

OER welcomes responses describing capabilities for the following desired variables/sensor types that may test the boundaries of the current AUV market.

Exploration Variable of Interest	Potential Sensing Approaches for AUVs
Biomass, density, distribution, and abundance: Microbes, plankton, meio- and macrofauna	Biological sampling (e.g. corals/sponge physical sample collection)
Geology/mineral resource characterization	Geological physical sample collection or in-situ mineral characterization

3. Requested Information

Respondents are requested to provide detailed information covering the following technical and operational areas as described by NOAA Ocean Exploration in this section. Please provide relevant specifications and supporting documentation, such as the full technical datasheet(s) for an AUV. <u>Example questions are provided to clarify the types of information that could be included but respondents can provide more or fewer details or answer some or all questions at their discretion</u>.

3.1. AUV System Specifications

3.1.1. Primary specification

- What is the AUV maximum **depth rating**?
- What is the system's endurance? What are **typical mission profiles** (survey altitude, line spacing, line holding ability in currents) used to quote endurance?
- What is the maximum operational speed? How does **endurance vary at** different speeds?
- What is the payload volume?

3.1.2. Battery specification

- Battery chemistry, capacity (kWh), and **usable** capacity?
- What is the battery **lifecycle** (cycles to X% capacity) and replacement lead time/cost?
- Charging:
  - Charge time, charger power requirements, cooling needs
  - What's the **turnaround time** in real ops?
- How are batteries packaged for shipping? Any compliance requirements?
- What shipboard safety controls are required (fire suppression guidance, storage temperature limits)?

3.1.3. Navigation, communications, autonomy, and recovery

- Navigation:
  - What is the **navigation accuracy** you can achieve with and without external aiding?
  - Navigation suite options: INS grade, DVL, pressure, USBL/LBL compatibility, use of satellite derived bathymetry as reference surface
  - What is the expected **drift** over time with your configuration?
  - To what extent do vehicles require command and control or positioning support from a ship? Can the vehicles be teamed with uncrewed surface vessels for command and control and/or positioning support (e.g. USBL, mounted on either USV or 'upward looking' from AUV) without similar communications and positioning from a ship? Has this been demonstrated? Would such a model impact staffing and ship berth requirements?
- Communications:
  - Acoustic modem options (rates/range), line-of-sight constraints?
  - Describe RF/Wi-Fi, satcom, or other options when surfaced?
- Autonomy:
  - Describe any capabilities with obstacle avoidance? Terrain following?
  - Lost-comm behaviors: what is the **fail-safes** and how configurable are they?
  - Autonomous behaviors/adaptive sampling: Please describe the AUVs ability to autonomously control its behavior, including ability to adaptively sample or replan missions based on autonomous behaviors as well as ability to add or modify such behaviors after delivery.
- Recovery aids:
  - Strobes, AIS, pingers, homing beacon options
  - What is the documented **lost vehicle** rate in comparable ops, and mitigations?

3.1.4. Software, certifications, and regulatory constraints

- Software safety practices: version control, change logs, rollback procedures?
- Export controls: any **ITAR/EAR** constraints on sensors, INS, autonomy, acoustics?
- Software update policy: software releases, compatibility guarantees, licensing model?
- Cybersecurity:
  - Authentication, encryption, removable media policies?
  - Logging/auditability, vulnerability disclosure policy?

3.2. Sensor Payload Capability

The AUV must accommodate a suite of scientific sensors. Please list all COTS sensors that can be integrated in accordance with Table 1 and 2. For each sensor please provide details:

- What **payload capacity** (mass, volume, power, data) is available now vs. "with upgrades"?
- What sensors are ready to be incorporated into the sensor payload?
- What **models** are supported and what are the performance specs **in AUV configuration**?
- Mounting: internal vs external, fairings, hydrodynamic penalties, pressure housing rating
- Interference concerns: acoustic cross-talk, EM noise, flow disturbance, vehicle self-noise
- What is the **payload swap time** and what tools/spares are required? Describe sensor modularity and upgradability.

3.3. Operations and Logistics

Please describe the operational requirements for the AUV.

3.3.1. Host vessel requirements

- Minimum vessel capabilities:
  - Deck area footprint (LARS + cradle + container + work area)?
  - Deck load limits (static + dynamic)?
  - Required **crane/A-frame SWL** and outreach; winch requirements; heave compensation needed?
  - Crew size to operate safely (AUV team + deck crew roles)
- **Launch & Recovery System (LARS)** options: A-frame, crane, davit, stern ramp; what are launch and recovery system options, space and power requirements, and limitations?
- Operational envelopes:
  - Minimum/maximum **freeboard** the LARS can handle
  - Minimum **water depth/clearance** needed for launch and recovery
  - Max sea state/wind for safe LARS operations (launch vs. recovery may differ)
  - What are the **limits**: max current, temperature range, salinity limits?
  - Can and have these systems be operated for at least 180 days a year at sea? What conditions (including staff berthing and in-season maintenance) are needed for that frequency of operations?
- Ship interfaces:
  - Power needs (voltage/phase/amps), grounding, shore-power compatibility
  - Network needs (ethernet/fiber), time-sync (NTP/PTP), GPS feed, optional ship INS integration
  - Containerization: Do you deliver as **ISO container labs**? If yes, size/weight/power requirements
- Location restrictions:
  - Can these systems be deployed in any location? Are there any geographic related limitations?

3.3.2. Maintenance, Warranty, Support and Training

- Maintenance / Support:
  - What are approximate maintenance hours at pre/post sortie, 50/100/500 hours, annual / depot?
  - Describe requirements for in-field repairs and spares stowage.
  - Estimated mean time between failures (MTBF) and required preventative maintenance schedule.
  - Do maintenance staff need ship berthing and can any work be done from shore?
  - What spares package is recommended for field ops (and cost)?

  - What are typical failure modes and what onboard health monitoring exists?
  - If selling hardware (not a CO-CO system), describe typical field support: on-site reps, remote support hours, response SLAs
  - Location of primary service and support centers?
- Warranty:
  - Warranty terms (what's excluded), and uptime guarantees if relevant
- Training
  - For government operated approaches, describe required training for: operator, maintainer, and mission commander tracks; duration; proficiency criteria
  - What documentation is delivered? operator manual, maintenance manual, wiring diagrams, parts lists, software manuals

3.4. Data Requirements

Respondents are requested to describe their capabilities to deliver data under the following scenarios:

- **Post-processed data** in commonly applicable formats (e.g. NetCDF, KVAL, 4K video etc.). Respondents should consult the NOAA National Centers for Environmental Information for data specifications and applicable ingest workflows/portals.
- **Raw or partially processed data.** Alternately, respondents are encouraged to describe workflows and pricing for this option.
- Time-tagging and navigation merge: how are nav and sensor data fused and exported?

For Respondents seeking to sell systems to NOAA (i.e. GO-GO or GO-CO business models), please provide descriptions and general costing for data management components, including software, and training in system data management that NOAA must have to operate the platform successfully.

3.5 Cost

DISCLAIMER: The Government is seeking Rough Order of Magnitude (ROM) pricing for market research purposes only. This is not a request for a formal proposal or a quote. Responses will be used solely for budgetary and planning purposes.

- What contracting model is available to access this system (e.g., GO/GO, GO/CO, CO/CO)?
- Please provide pricing information for these models to help determine the best business model for future missions:
  - Estimated system acquisition cost (base price, including essential launch/recovery equipment).
  - Estimated cost for a three-year comprehensive support and maintenance agreement.
  - For CO/CO systems, estimated daily operating costs and mobilization costs.